



HAZOP Leadership training for GC

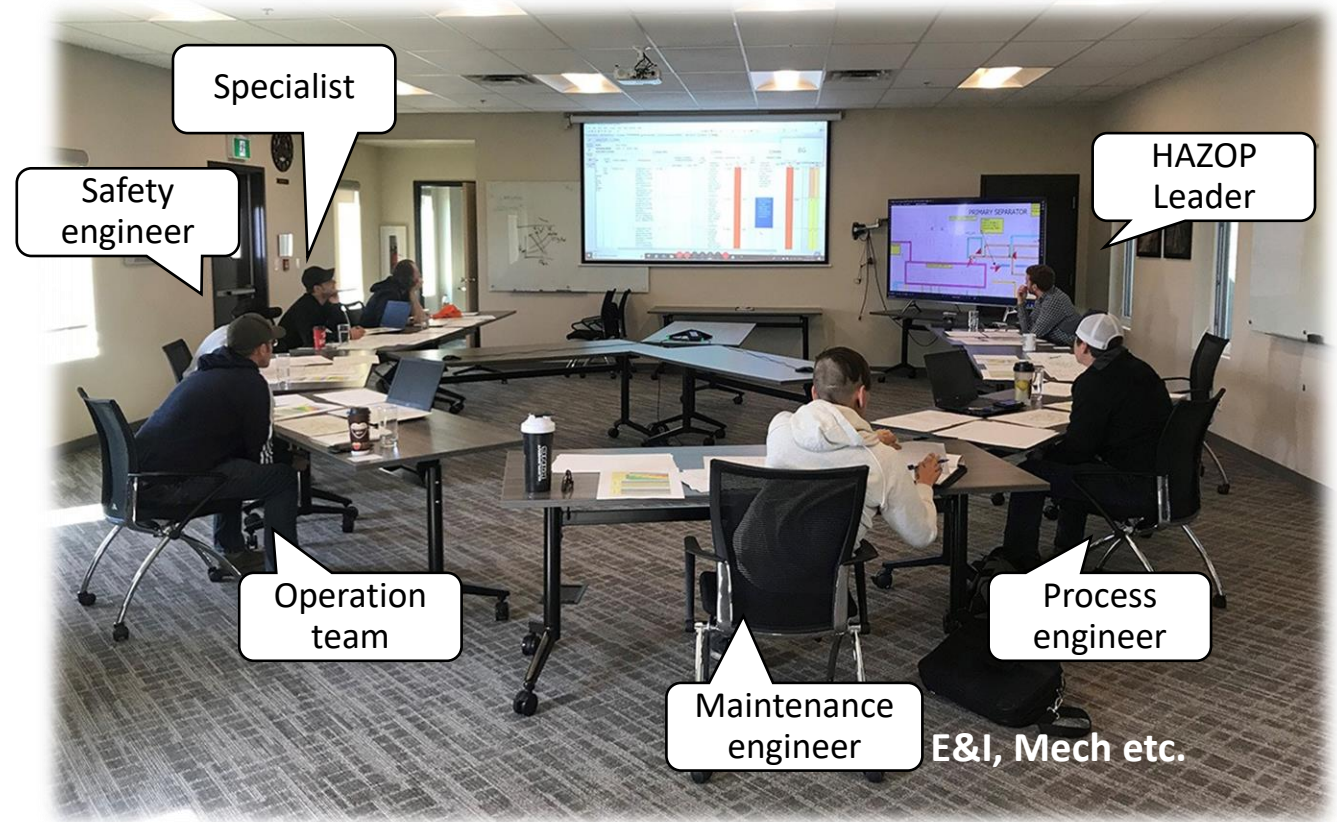
Chapter 1 - Hazard and Risk concept

Technical Safety Service Division (Q-TS-TS), 1 - 3 November 2021

Activity-1 : What are your position? (Menti-1 Check-in)

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<https://www.menti.com/1g4hx55ryn>



Content

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- ☐ Hazard, Harm & Risk
- ☐ Risk assessment process
- ☐ Acceptable risk



How many Major Accident Events (MAEs) per year do you think happen?

Global Major Accident Events 2019 – 2020 (Jan – Apr)

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How many Major Accident Events (MAEs) do you think happen monthly?

Global Major Accident Events 2019



Global Major Accident Events 2020 (Jan – Apr)



Global Major Accident Events 2019 – 2020 (Jan – Apr)

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1



14-Jan-20: Spain : Fire & Explosion at Ethylene Oxide (2 fatalities & 8 injuries)

2



24-Jan-20: USA : Explosion at Propylene Tank (2 fatalities & 2 injuries)

3



12-Feb-20: USA : Fire & Explosion at Refinery (16 Tanks & Process damaged)

4



14-Feb-20: China : Fire & Explosion at Chemical Factory (5 fatalities & 10 injuries)

5



16-Feb-20: Pakistan : Methyl Bromide & H₂S Leakage (14 fatalities & >100 injuries)

6



26-Feb-20: USA : Methyl Bromide & H₂S Leakage (14 fatalities & >100 injuries)

Global Major Accident Events 2019 – 2020 (Jan – Apr)

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7



4-Mar-20: South Korea : Fire & Explosion at Chemical Plant (many injuries)

8



15-Mar-20: Nigeria : Pipeline & Tanker Truck Explosion (15 fatalities & 25 injuries)

9



15-Mar-20: Malaysia : Fire & Explosion at Diesel Hydrotreater Unit (5 fatalities)

10



10-Apr-20: USA : Fire & Explosion at Refinery (1 injury)

11



13-Apr-20: India : Fire & Explosion at Chemical Plant (2 fatalities & 1 injury)

12



23-Apr-20: Taiwan : Ethylene Dichloride Leakage (Single fatality)

HAZARD

VS

RISK

A **HAZARD** is something
that has the potential
to harm you



RISK is the likelihood
of a hazard
causing harm



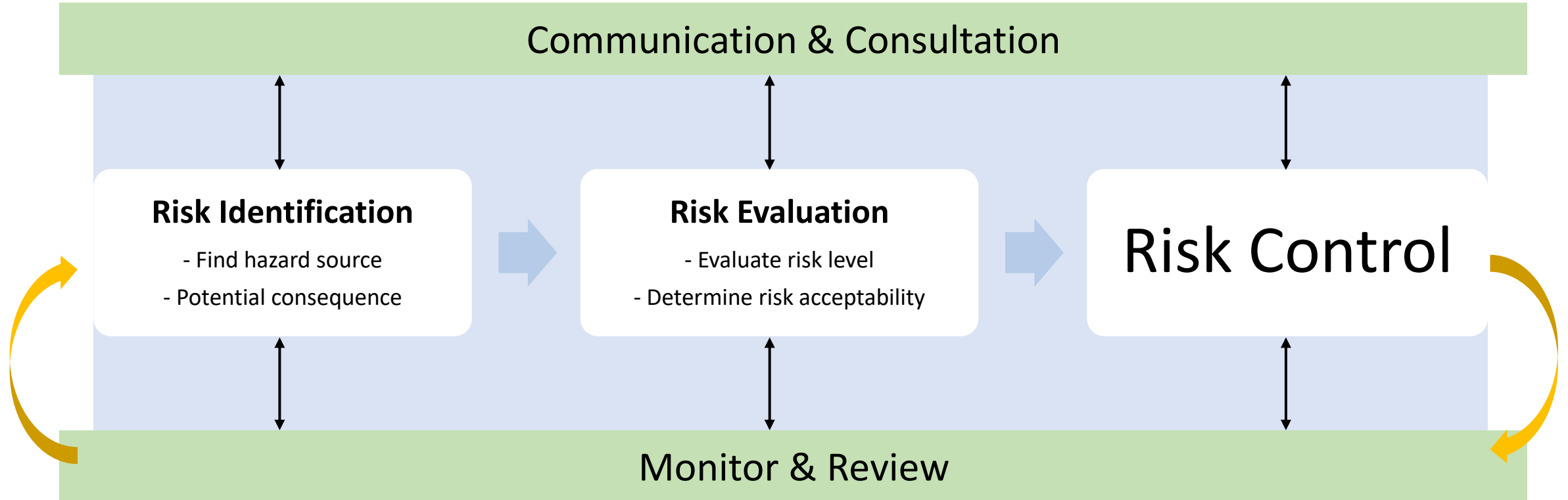
How to manage risk?

- ❑ **Hazard** is a physical or chemical condition that has the potential for causing harm to people, property, or the environment. Hazards are intrinsic to a material, or its conditions of use.
- ❑ **Harm** is defined as injury or damage to the health of people, or damage to property/asset or the environment
- ❑ **Risk** is defined as the product of probability and consequence when probability and consequence are expressed numerically.

$$\text{RISK} = (\text{Frequency}) \times (\text{Consequence})$$

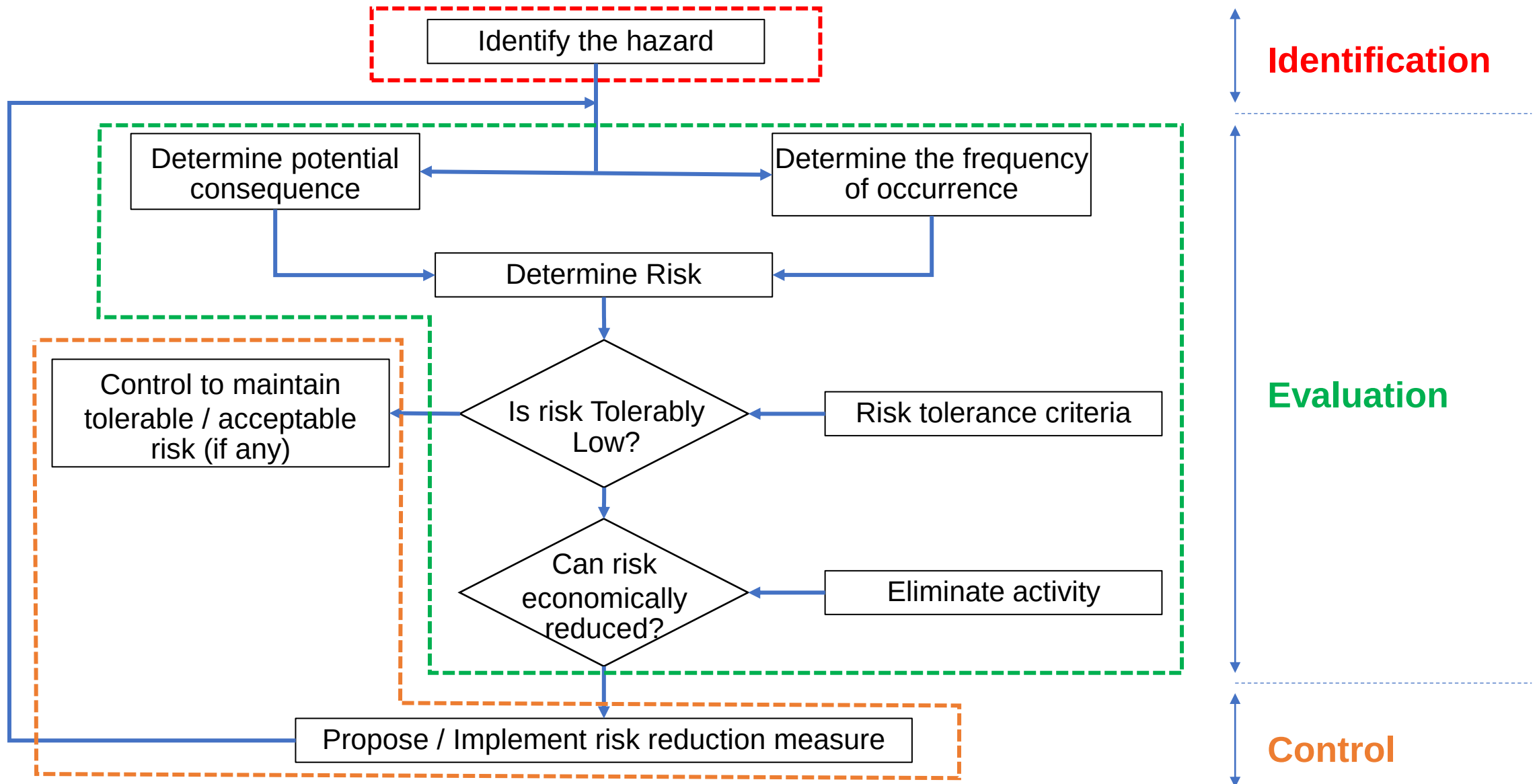
Three (3) types of risk determination

- Qualitative : Risk matrix
- Semi-Quantitative : Layer of protection Analysis
- Quantitative : Quantitative Risk Assessment (QRA)



Risk assessment process

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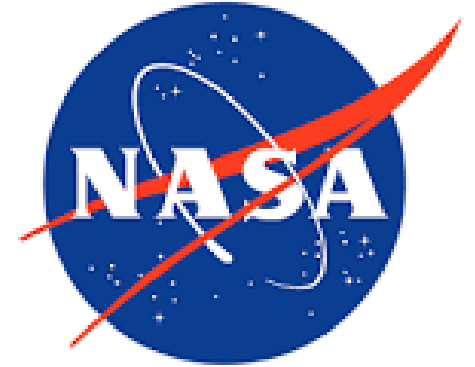
Acceptable risk

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Quote from NASA senior scientist regarding the Saturn 5 rocket, from first Apollo mission.

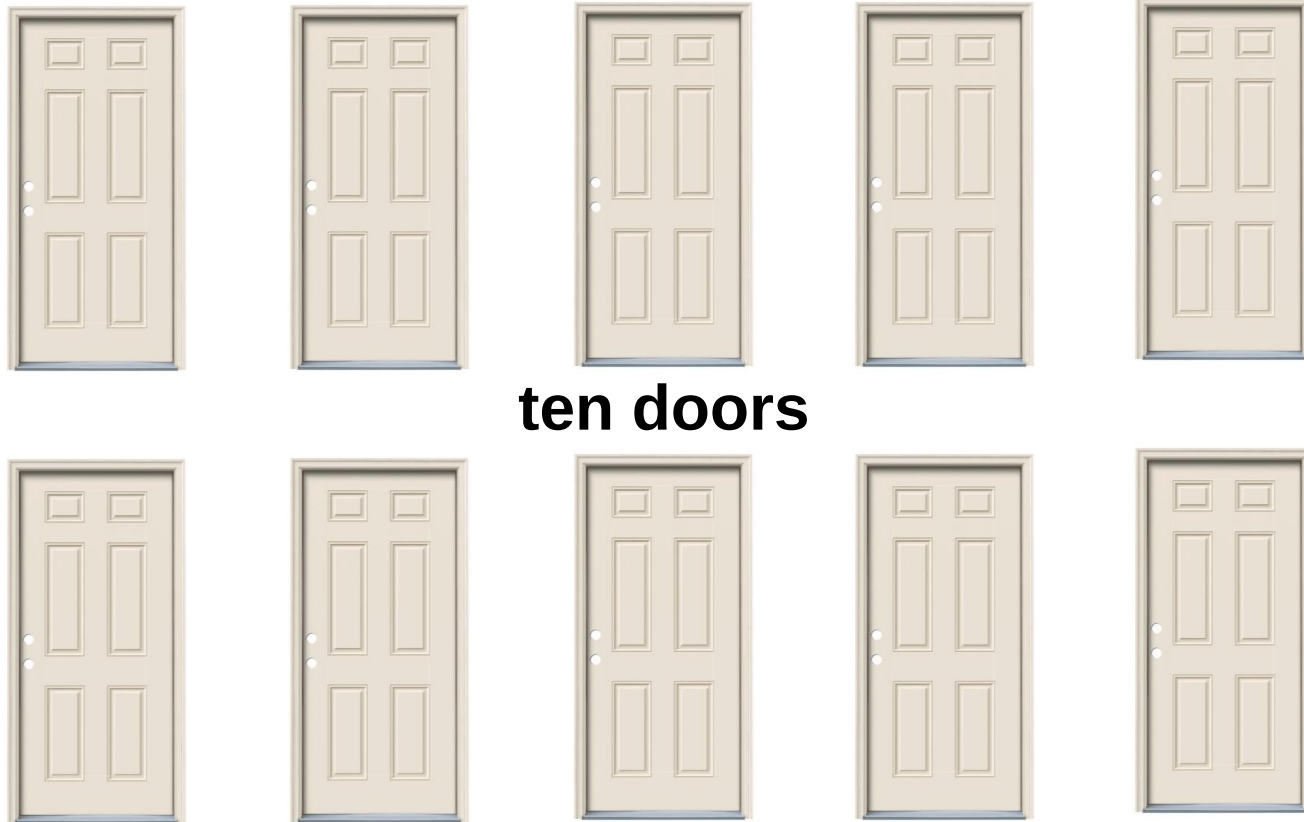
"You want a valve that doesn't leak, and you try everything possible to develop one.

**But the world provide you with a leaking valve.
You have to determine how much leakage you can tolerate"**



Acceptable risk

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ten doors



$\$1,000,000 \times 9$



LHMEL x 1
(large hungry man-eating lion)

Would you accept this Risk?

Acceptable risk

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\$1,000,000 x 999

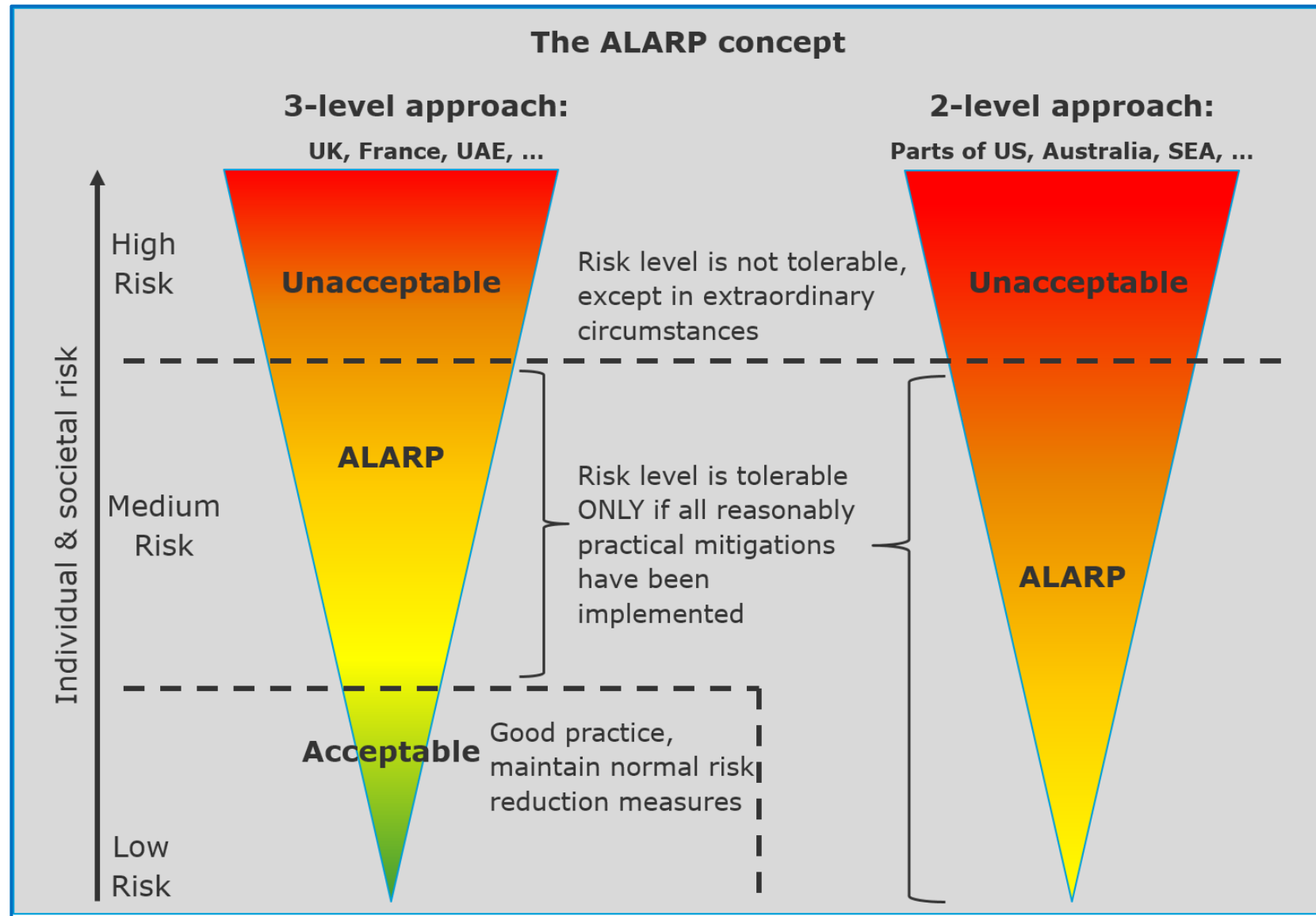


LHMEL x 1
(large hungry man-eating lion)

Well – what about this!

Acceptable risk

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GC PHA Risk matrix

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W-(Q-TS)-002: Operational Risk Assessment Matrix and Its Applications

Likelihood	General Term (No.)	How often?	Hazard Severity Rating				
	Frequent (5)	Has happened more than once per year in the Location	Low	Medium	High	Extreme	Extreme
	Likely (4)	Has happened at the Location or more than once per year in the PTTGC group	Low	Medium	High	High	Extreme
	Possible (3)	Possible to happen in the PTTGC group or more than once / year in the Industry	Low	Low	Medium	High	High
	Unlikely (2)	Possible to occur in the Industry (or has occurred)	Very Low	Low	Low	Medium	Medium
	Improbable (1)	Unlikely to occur in the Industry (or has not occurred)	Very Low	Very Low	Low	Low	Low
Consequence	Severity Number		(1)	(2)	(3)	(4)	(5)
	People (Safety, Health, Morale)		- No injury or First Aid case - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike
	Environment		No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
	Economic (Total Loss)	GPC (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		BU (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
		Small BU (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	≥ 10 M
	Social (Community, Reputation, Customer, Law/Regulation)		- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company

GC Master Risk matrix (People)

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Consequence on **People** includes the effect to Safety, Health and Morale

Level (Scaling no.)	Definition
Very low (1)	No /Slight effect – not affecting work performance and not affecting daily life activities, Examples: • No injury or First aid case. • No or Very low health effect; o exposure to health hazards that give rise to noticeable discomfort, minor irritation or transient effects reversible after exposure stops. • No or Minimal morale impact
Low (2)	Minor effect – affecting work performance in the short term, Examples: • Medical treatment or Restriction to work. • Low health effect; o illnesses such as skin irritation or food poisoning. • A minor unplanned or uncontrolled release (loss of primary containment : LOPC) of hazardous material, from a process that results a release less than the threshold quantities described in Table 2 of “ANSI/API RECOMMENDED PRACTICE 754” in anyone-hour period. • A small fire or explosion. • Short-term morale impact
Medium (3)	Moderate effect – affecting work performance in the longer term, Examples: • Loss time injury • Long term disabilities. • Medium health effect; - o illnesses such as sensitization, noise induced hearing loss, chronic back injury, repetitive strain injury or stress. • An unplanned or uncontrolled release (loss of primary containment : LOPC) of hazardous material, from a process that results a release greater than the threshold quantities described in Table 2 (but less than threshold in Table 1) of “ANSI/API RECOMMENDED PRACTICE 754” in anyone-hour period. • Long-term morale impact
High (4)	Major effect – resulting from injury, occupational illness or morale, Examples: • Single fatality or Permanent total disabilities. • High health effect; o illnesses such as corrosive burns, asbestosis, silicosis, cancer and serious work related depression. • An unplanned or uncontrolled release (loss of primary containment : LOPC) of hazardous material, from a process that results a release greater than the threshold quantities described in Table 1 of “ANSI/API RECOMMENDED PRACTICE 754” in any one-hour period. • Protesters rally or official complaint
Extreme (5)	Massive effect – resulting from injury, occupational illness or morale. Examples: • More than one fatality • Major fire or explosion resulting in more than 1 fatalities. • Very high health effect. • Multiple asbestosis cases traced to a single exposure situation - o Cancer to a large exposed population • Employees or Contractors strike

GC Master Risk matrix (Environmental)

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The bullet points in the environmental effect table are a mixture of

- Effects, e.g. groundwater contamination.
- Events with the potential for environmental effect, e.g. exceeding a limit.
- Indicators of potential effects, e.g. complaints

Level (Scaling no.)	Definition
Very low (1)	No/Slight effect: Minor environmental damage – contained within the premises. Example: • Small spill in process area or tank farm area that readily evaporates
Low (2)	Minor effect: Minor environmental damage, but no lasting effect. Example: • Small spill off-site that seeps into the ground. • On-site groundwater contamination. • Complaints from community up to 10 individuals in the same case. • Single exceeding of statutory or other prescribed limit. • Spill any loss of containment of hydrocarbon and non-hydrocarbon of not more than 1 barrel or 159 Liters that reaches the environment, irrespective of quantity recovered
Medium (3)	Moderate effect: Limited environmental damage that will persist or require cleaning up. Example: Loss time injury • Spill into soil/sand that requires removal and disposal of a large quantity of soil/sand. • Observed off-site effects or damage, e.g. fish kill or damaged vegetation. • Off-site groundwater contamination. • Complaints from community organizations (or more than 10 individuals in the same case). • Frequent exceeding of statutory or other prescribed limit, with potential long term effect. • Spill any loss of containment of hydrocarbon and non-hydrocarbon of greater than 1 barrel or 159 Liters that reaches the environment, irrespective of quantity recovered.
High (4)	Major effect: Severe environmental damage that will require extensive measures to restore beneficial uses of the environment. Example • Oil spill at a jetty during tanker (Off) loading that ends up on local beaches, requiring clean-up operations. • Off-site groundwater contamination over an extensive area. • Many complaints from community organization or local authorities/government. • Extended exceeding of statutory or other prescribed limits, with potential long term effects. • Spill any loss of containment of hydrocarbon and non-hydrocarbon of greater than 50 barrel or 7950 Liters that reaches the environment, irrespective of quantity recovered
Extreme (5)	Massive effect: Persistent severe environmental damage that will lead to loss of commercial, recreational use or loss of natural resources over a wide area. Example: • Spillage resulting in pollution of a large part of a river estuary/the sea and extensive clean-up and remediation measures. • Significant spill: ○ Any hydrocarbon or Any non-hydrocarbon spill of greater than 100 barrels or 15900 Liters released into the environment; ○ Any hydrocarbon or Any non-hydrocarbon spill released into a sensitive environment ○ Any hydrocarbon or Any non-hydrocarbon spill is reported in the organization's financial statement. (1 barrel = 159 Liters, Based on assumption that barrel of oil and chemicals is the same quantity.)

GC Master Risk matrix (Economic)

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Economic Consequence is the total financial loss that includes

- Product loss (It consider Product Loss, Flare Loss or Product Off-spect, which Product price consider as average last year Price of Product)
- Opportunity loss (It consider Margins Loss , which Margins price consider as average last year Margins)
- Asset damage loss
- Clean up Cost

Level (Scaling no.)	Definition
Very low (1)	No/Slight damage • Upstream Plant: Costs < 0.3 M THB • Downstream Plant: Costs < 0.1 M THB • GC-S and smaller Plant: Costs < 0.01 M THB
Low (2)	Minor damage • Upstream Plant: Costs: Costs 0.3 -<3 M THB • Downstream Plant: Costs 0.1 -<1 M THB • GC-S and smaller Plant: Costs 0.01 -< 0.1 M THB
Medium (3)	Moderate damage • Upstream Plant: Costs: Costs 3 -< 1 M THB • Downstream Plant: Costs 1 -<10 M THB. • GC-S and smaller Plant: Costs 0.1 -< 1 M THB
High (4)	Major damage • Upstream Plant: Costs: Costs 30 -<300 M THB • Downstream Plant: Costs 10 -<100 M THB. • GC-S and smaller Plant: Costs 1 -< 10 M THB
Extreme (5)	Massive damage • Upstream Plant: Costs: Costs ≥ 300 M THB • Downstream Plant: Costs ≥ 100 M THB. • GC-S and smaller Plant: Costs ≥ 10 M THB

- Upstream Plant : O-P1, O-P2, O-P3, U-P1, U-CM, R-P1, R-RM, A-P1, A-P2
- Downstream Plant : P-HD1, P-HD1, P-LD1, P-LL1, P-LL2, PH-P1, PH-P2, EO/EG, EA, GGC, Vencorex
- GC-S and smaller Plant : GCS

GC Master Risk matrix (Social)

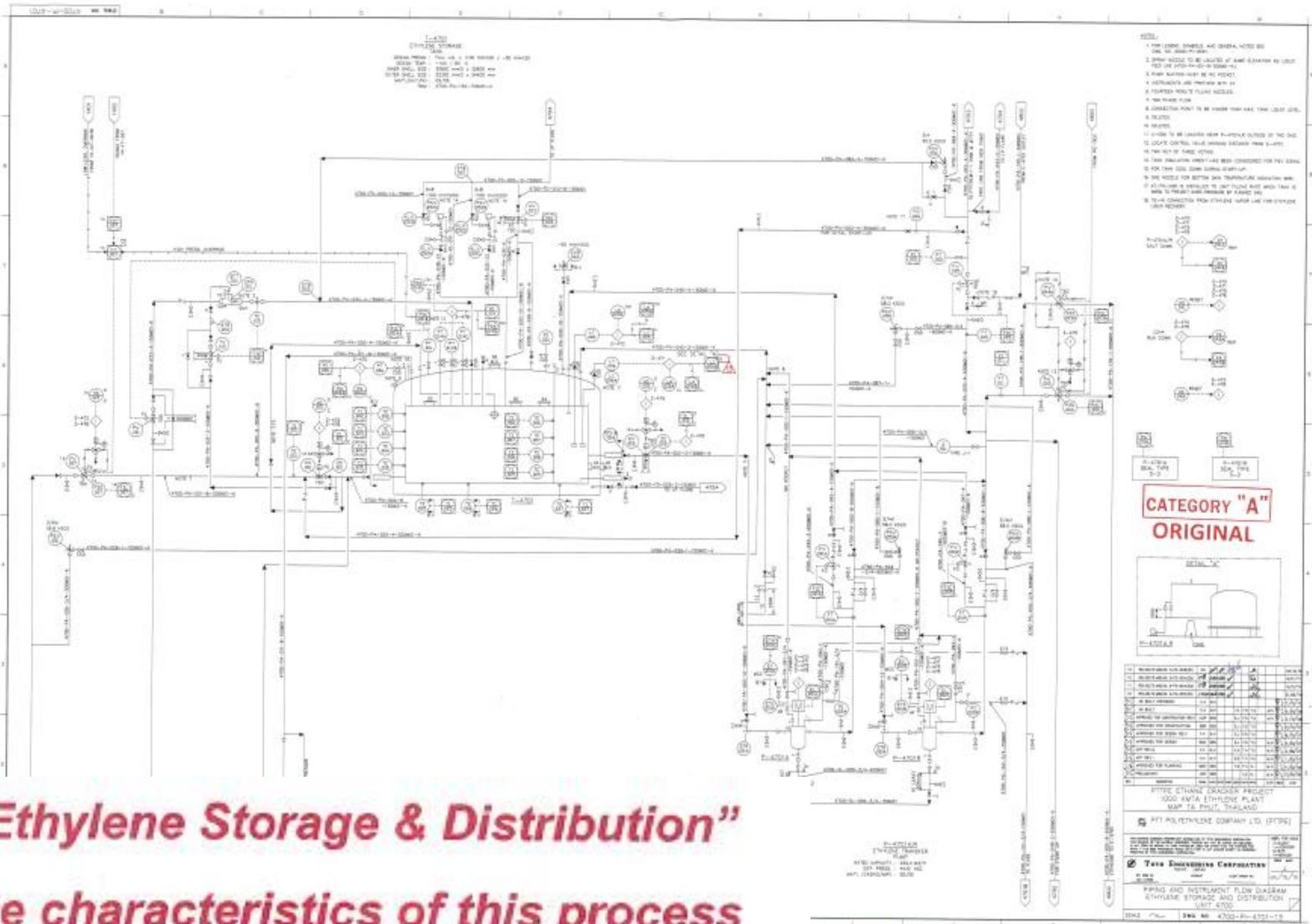
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Social Consequence includes the effect to Community, Reputation, Customer and Law & Regulation

Level (Scaling no.)	Definition
Very low (1)	No/Slight damage - Local public awareness but no discernible concern. - No Media coverage. - No/Slight impact to customer. - No fault or insignificant fault (of complying with laws/articles of association) though within acceptable circumstances.
Low (2)	Minor damage - Minor impact to the community with short term recovery. - Normal community functioning with some inconvenience. - Local public concern. - Local media coverage. - Verbal complaints by customer - Partly comply with laws/articles of association, however, the contradiction are within manageable circumstances and/or may be subject to warning/fine
Medium (3)	Moderate impact –Significant impact in region or country - Moderate impact to the community with long term recovery. - Public protective measures (e.g. road closure). - Regional public concern. - Local stakeholders, e.g. community, NGO, industry and government, are aware. - Extensive attention in local media. Some regional media coverage. - Official letter complaint from customer. - Non-compliance with laws/articles of association and being brought action against.
High (4)	Major impact – likely to escalate and affect Group reputation - An officially declared community evacuation or community shelter-in-place. - Major impact with national public concern. - Impact on local and national stakeholder relations. National government and NGO involvement with potential for international NGO action. - Extensive attention in national media. Some international coverage. - Customer less purchase. - Violate the laws/ articles of association and/or subject to temporary suspension order to operate
Extreme (5)	Massive impact –Severe impact on Group reputation - Massive impact with International public concern. - Serious damage to infrastructure causing significant disruption to, or loss of, key services for prolonged period. Community unable to function without significant support. - High level of concern amongst governments and action by international NGOs. - International media attention. - Customer stops purchase. - Violate the laws/ articles of association, subject to suspension order to operate until further decree otherwise by court and/or subject to order to dissolve the company.

Activity-2 : Identify the hazard (Menti-2)

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T-4701
ETHYLENE STORAGE
TANK

DESIGN PRESS. : FULL LIQ. + 1100 mmH2O / -50 mmH2O
DESIGN TEMP. : -104 / 65 °C
INNER SHELL SIZE : 30800 mmID x 32600 mm
OUTER SHELL SIZE : 32300 mmID x 34400 mm
MAT'L(OUT/IN) : CS/SS
TRIM : 4700-PA-130-150M01-K

<https://www.menti.com/1g4hx55ryn>



“Ethylene Storage & Distribution”
the characteristics of this process
will be explained first

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End of Chapter 1: Hazard and Risk concept
